Newton's Second Law

Exam code: 0625 & 0972

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Newton's second law

- Newton's **second law of motion** states:

The acceleration of an object is proportional to the resultant force acting on it and inversely proportional to the object's mass

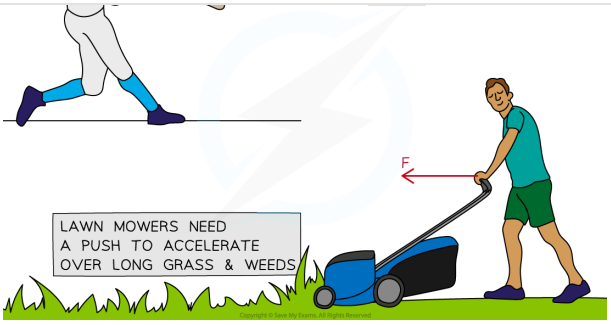
- Newton's second law of motion explains what happens when a **non-zero resultant force** acts on an object
- A resultant force occurs when the forces acting on an object are **not balanced**
- A resultant force acting on an object will cause a **change in the object's motion**
- This change in motion is an **acceleration**:
  - Speeding up
  - Slowing down
  - Changing direction

- If the resultant force on an object is not zero, the object will **accelerate** in the **direction** of the **resultant force**
- The **bigger** this resultant force, the **larger** the acceleration
- For a given force, the **greater** the object's mass, the **smaller** the acceleration experienced

Examples of Newton's second law

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Objects like baseballs and lawnmowers accelerate when a resultant force is applied on them. The size of the acceleration is proportional to the size of the resultant force

Calculations using Newton's second law

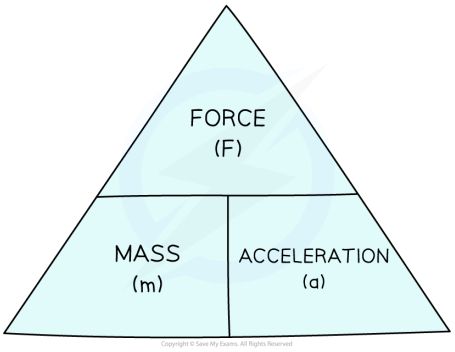
Extended tier only

- Newton's second law can be expressed as an equation:

$$F = ma$$

- Where:
  - F = resultant force on the object, measured in newtons (N)
  - m = mass of the object, measured in kilograms (kg)
  - a = acceleration of the object, measured in metres per second squared (m/s^2)
- The **acceleration** occurs in the **same direction** as the **resultant force**

Formula triangle for acceleration, mass and resultant force



To use a formula triangle, simply cover up the quantity you wish calculate and the structure of the equation is revealed

- A more detailed explanation of how to use formula triangles is covered in the revision note Speed and velocity

Worked Example

A car salesman says that their best car has a mass of 900 kg and can accelerate from 0 to 27 m/s in 3 seconds.

Calculate:

a) the acceleration of the car in the first 3 seconds.

b) the force required to produce this acceleration.

Answer:

Part (a)

Step 1: List the known quantities

- Initial velocity, $v = 0 \text{ m/s}$
- Final velocity, $u = 27 \text{ m/s}$
- Time, $t = 3 \text{ s}$

Step 2: State the equation for acceleration, in terms of change in velocity

$$a = \frac{v - u}{t}$$

$$a = \frac{27 - 0}{3}$$

$$a = 9 \text{ m/s}^2$$

Part (b)

Step 1: List the known quantities

- Mass of the car, $m = 900 \text{ kg}$
- Acceleration, $a = 9 \text{ m/s}^2$

Step 2: Identify which law of motion to apply

- The question involves quantities of **force**, **mass** and **acceleration**, so Newton's second law is required:

$$F = ma$$

Step 3: Calculate the force required to accelerate the car

$$F = 900 \times 9$$

$$F = 8100 \text{ N}$$

Worked Example

Three shopping trolleys, **A**, **B** and **C**, are being pushed using the same force. This force causes each trolley to accelerate.

State which trolley would have the smallest acceleration. Explain your answer.

Answer: C

Step 1: Identify which law of motion to apply

- The question involves quantities of **force** and **acceleration**, and the image shows trolleys of different **masses**, so **Newton's second law** is required:

$$F = ma$$

Step 2: Re-arrange the equation to make acceleration the subject

$$a = \frac{F}{m}$$

Step 3: Explain the inverse proportionality between acceleration and mass

- Acceleration is **inversely proportional** to mass
- This means that for the **same amount of force**, a **large mass** will experience a **small acceleration**
- Therefore, trolley **C** will have the smallest acceleration because it has the largest mass

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